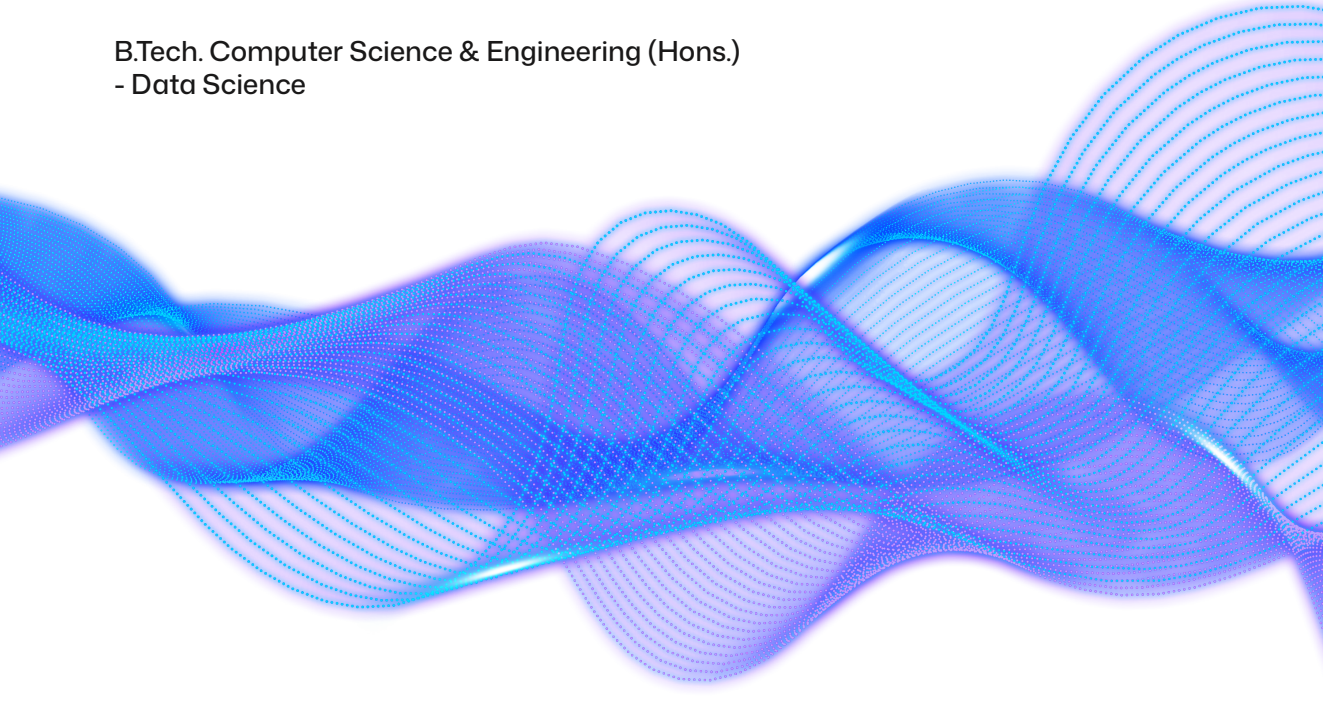
VIDYASHILP UNIVERSITY

Experiential Program in

DATA SCIENCE

B.Tech. Computer Science & Engineering (Hons.)
- Data Science



VIDYASHILP
SCHOOL OF ENGINEERING & TECHNOLOGY

About the Program

Shaping Real-World Data Scientists
to Drive Impactful Industry Solutions.

B.Tech. (Hons.)
in Computer Science & Engineering - Data Science

Real-World Data Scientists

- Cloud Computing & AI Model Deployment
- Agentic AI
- Reinforcement Learning
- Multimodal Learning

Data & Technology

- Bayesian Methods in Machine Learning / Generative AI
- Data & Computer Technology
- Cloud. IoT, Networking

Future-Ready Industry Applications

- MLOps
- Big Data Analytics
- Big Data Frameworks
- Full Stack Development

Lead Global Innovation with World-Class Faculty

Our distinguished faculty have advanced academic training from some of the world's top universities, including IISc - Bengaluru, IIT Bombay, BITS - Pilani, IIT - Dhanbad, IIM - Kozhikode, Bharati Vidyapeeth University- Pune. Learn from their expertise, collaborate on groundbreaking research, and embark on your own journey of educational transformation.

Why Choose the B.Tech. (Hons.) in Computer Science & Engineering Data Science Program at VU?

Our B.Tech. (Hons.) in Computer Science & Engineering - Data Science program is thoughtfully crafted to integrate core computer science principles with advanced data science techniques, equipping students to tackle complex real-world challenges.

Cutting-Edge Curriculum: Integrates Computer Science and Engineering with advanced Data Science topics like Artificial Intelligence, Machine Learning, and Deep Learning.

Specialization Tracks: Focus areas such as Applied Deep Learning and Full Stack Development, allowing students to tailor their learning.

Interdisciplinary Opportunities: Minors in Psychology, Design, Marketing, Finance, Law, Start-ups, and Economics broaden career prospects.

Holistic Skill Development: Emphasis on soft skills such as communication, critical thinking, and teamwork.

Hands-On Learning: Capstone projects, internships, and practical sessions ensure industry readiness.

Future-Ready Graduates: Prepares students for roles in AI, data-driven decision-making, and innovative technology solutions.

Expert Faculty: Learn from accomplished academicians and industry professionals with expertise in computational sciences and cutting-edge technologies.

State-of-the-Art Infrastructure: Access to modern labs and tools for practical and research work.

Global Perspective: Curriculum designed to address global challenges and equip students for global careers.

Course Distribution Across the Program

The undergraduate journey of a data science student at Vidyashilp University is meticulously crafted to integrate the University Core, Program Foundation, and Major discipline courses, fostering a transformative and enriching educational experience.

University Core

The **University Core** curriculum equips students with critical thinking, problem-solving, and leadership skills, fostering responsible global citizens.

Program Foundation

The **Program Foundation** courses establish a strong base in mathematics, computational thinking, and design, essential for advanced data science studies.

Program Core

The **Program Core** focuses on practical and rigorous learning across various disciplines of computer science, including Artificial Intelligence and Internet of Things, shaping future-ready graduates.

Program Honors

The **Program Honors** offers scholarly rigor in key areas such as Machine Learning, Deep Learning, and Cloud Computing, equipping students with expertise in cutting-edge technologies.

Specialization Track / Electives

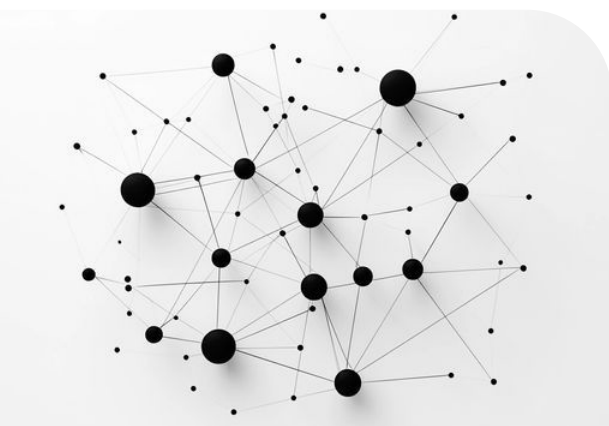
A key highlight of the program is the **Specialization Tracks**, allowing students to focus on areas such as Applied Deep Learning or Full Stack Development to align with their career aspirations.

Minor Discipline

Students can pursue **Minors** in diverse domains, fostering interdisciplinary knowledge and innovation.

Research and Internship

Students also undertake **Internships** with leading companies gaining valuable industry exposure and practical experience.



Overview of the B.Tech. Framework

Program Foundation

Materials for Smart Devices | Sustainable Smart Materials | Design Workshop | Calculus | Probability and Statistics | Linear Algebra | Advanced Statistical Analysis and Research Methods with R

Honors Program Core

Advanced Exploratory Data Analysis | Optimization Techniques for Data Science | Machine Learning | Deep Learning | Cloud Computing and AI Model Deployment

Specialization Tracks

Track 1:
Applied Deep Learning | Probabilistic Graph Models | Bayesian Methods in Machine Learning / Generative AI | Natural Language Processing | Multi-Modal Learning | Reinforcement Learning

Track 2:
Full Stack Development | Backend Systems and API Development | Big Data Analytics | Web Frameworks | Mobile Application Development | Performance Optimization and Testing

University Core

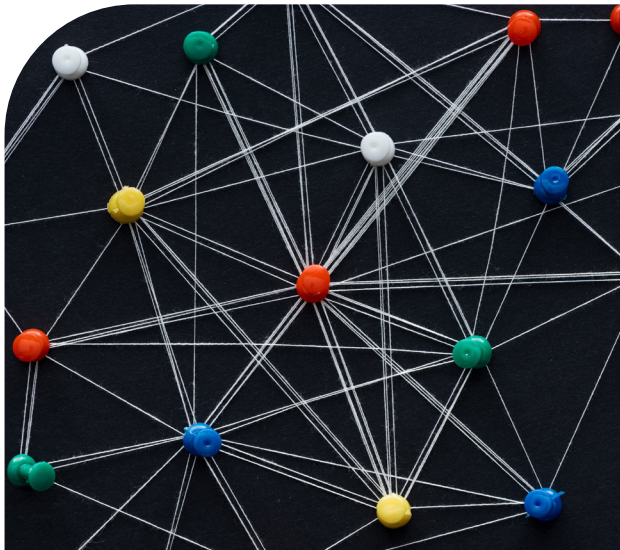
Communication Skills | Critical Thinking | Problem Solving for Design Thinking | Environmental Studies and Sustainability | Fundamentals of Indian Constitution and Human Rights | Community Engagement | Socio-Cultural Perspectives on Indian Life | Learning to Learn

Program Core

Data Structures | Theoretical Computer Science | Analysis of Algorithms | Digital Design and Computer Organization | Object-Oriented Programming | Microprocessors and Computer Architecture | Operating Systems | Networks | Databases Management | Programming in Python | Data Visualization and Storytelling | Artificial Intelligence | Internet of Things

Minors

Economics | Psychology | Design | Marketing | Finance | Law | Start-ups



Faculty

P S Sastry
Dean, School of Engineering & Technology
PhD, IISc, Bengaluru

M.S. Smitha Rao
Program Chair, B.Tech (Hons)- Computer Science & Engineering (Data Science)
PhD, Bharati Vidyapeeth University, Pune

Subrahmanya K N
Assistant Professor
School of Engineering & Technology
PhD, IIT Kanpur

D.C. Kiran
Associate Professor, School of Engineering & Technology
PhD in Computer Engineering, BITS Pilani

Mahendra B M
Associate Professor, School of Engineering & Technology
PhD in Computer Vision & Image Processing, Visvesvaraya Technological University

N. Mehala
Associate Professor, School of Engineering & Technology
PhD in Computer Science, BITS Pilani

Shabbeer Basha
Associate Professor, School of Engineering & Technology
PhD, Indian Institute of Information Technology Sri City (IIITS)

Shital D Bhatt
Associate Professor,
School of Engineering &
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PhD, Gujarat
Technological University

Anupama A.V.
Assistant Professor,
School of Engineering
& Technology
PhD in Material Science,
IISc, Bengaluru

Bijil Prakash
Assistant Professor,
School of Engineering
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PhD in Applied
Mathematics, BITS Pilani

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PhD in Computational
Biology, IMSc, Chennai

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PhD in Applied
Mathematics,
University of Delhi

Ravi Kant Rai
Assistant Professor,
School of Engineering
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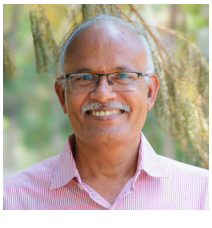
PhD, IIT Bombay



R.P. Suresh
Professor of Practice,
School of Engineering
& Technology
PhD in Statistics,
University of Pune

M. Mallikarjuna
Assistant Professor,
School of Engineering &
Technology
PhD in Computer Science
& Engineering - Pursuing
at IIT (Indian School of
Mines), Dhanbad

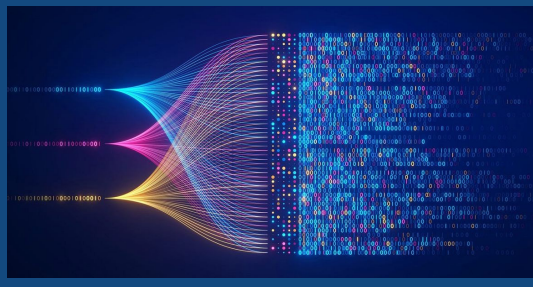
Saroj Snehal Shivagunde
Assistant Professor
School of Engineering
& Technology
PhD (CSE), IIT,
Guwahati

KNOW MORE ABOUT OUR FACULTY

Learning Journey Across Academic Years

Year 1: Foundation

Build a robust foundation in computer science and mathematics through courses like Data Structures, Calculus, and Linear Algebra. Develop essential analytical and problem-solving skills.



Year 2: Exploration

Explore advanced computational concepts and data science techniques, including Programming in Python, Probabilistic Graph Models, and Statistics. Strengthen your understanding of theoretical and applied computer science.

Year 3: Specialization

Dive deep into specialization tracks such as Applied Deep Learning or Full Stack Development, gaining expertise in advanced topics like Natural Language Processing, Big Data Analytics, and Cloud Computing.

Year 4: Proficiency

Master technical and interdisciplinary competencies through a capstone project (1 semester), industry internships, and courses like Internet of Things and AI Model Deployment.

Achievements

IEEE Student Chapter

1st place at the IEEE Leadership Summit 2024 held in Hyderabad

1st Prize in Hackathon

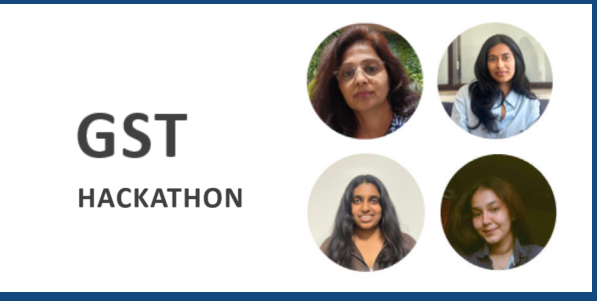
3rd Prize in Code 0 Contest

2nd Prize Best IEEE Student Branch

Bengaluru Mobility Challenge - Detect vehicle at every turning patterns and predict the number of vehicles 30 mins into future

Awarded 1st Place - 4 lakhs

Prof. Smitha Rao
Mr. Arnold Sachith (Research Associate)
Mr. Mohit Bansal



GST Hackathon

Online Challenge for Developing a Predictive Model in GST, March 2025

Prof. Smitha Rao
Shreya BM (Student, B.Tech. Hons.)
Olavu C (Student, B.Tech. Hons.)
Joshitha S (Student, B.Tech. Hons.)

5G Hackathon

Developed an AI/ML- based App which assists LEA's in investigation, evidence collection , and drafting of charge sheets.

Awarded Runner Up

Prof. Smitha Rao
Mr. Arnold Sachith
Mr. Mohit Bansal

Basha Techathon by Bhashini and Google Cloud

Voice to text and bucketing of complaints into correct type/sub types through AI/ML to help Indian Railways in faster processing of customer complaints

Prof. Smitha Rao
Mr. Arnold Sachith
Mr. Aman Kumar

Seamless Service: Fewer Clicks, More Care with Video-Driven Railway Complaint Management

Mr. Mohit Bansal
Mr. Amar Singh
Mr. Subhadeep Guchhait

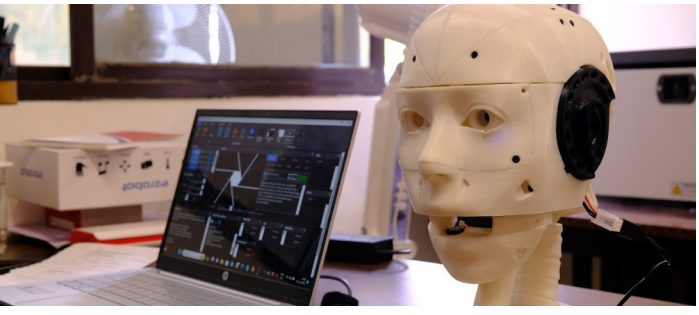
ISB | CyberPeace Foundation

Deep Fake Detection and Verification, also build a User-Friendly Deepfake Verification App

Prof. Smitha Rao
Mr. Arnold Sachith
Mr. Mohit Bansal

Shilpites Showcase

Data Science students at Vidyashilp University work on challenging projects that help them develop strong analytical skills and a deeper understanding of the domain. Their projects focus on solving real-world problems through data-driven decisions.



VURA

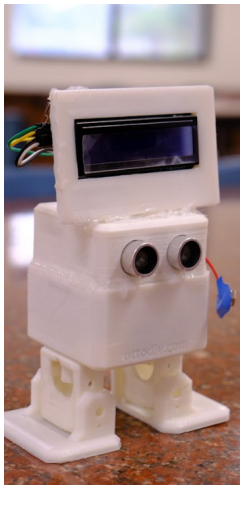
VURA combines the power of speech and image recognition, showcasing the innovative spirit of students at VU. By integrating AI, Machine Learning, and Natural Language Processing, it highlights how students apply cutting-edge technologies to create interactive, real-world solutions.

FORENSIC SKETCHING ROBOT

The AI-powered forensic sketching robot combines facial landmark detection, speech-to-text, and cloud computing to address the limitations of traditional forensic sketching. By providing an interactive, AI-enhanced system, it allows for more precise suspect identification, reducing human error and bias while ensuring greater forensic accuracy and credibility.

FORENSIC VISION

Leveraging computer vision and UV light, Forensic Vision enhances crime scene evidence detection by reducing human error and bias. It offers a secure, transparent process for evidence logging and filing, ensuring tamper-proof records and the legal admissibility of forensic data while improving the accuracy of investigations.



ULTRASONIC DISTANCE SENSOR

Redefining precision in measurement, the Ultrasonic Distance Sensor enhances applications in Autonomous Navigation, Healthcare Robotics, and Marine Research, ensuring accurate data for critical operations.

AUTOMATED FIREFIGHTER

Designed for rapid response and pinpoint accuracy, the Automated Firefighter is set to revolutionize firefighting and search and rescue missions, improving search and rescue in high risk environments.

Paper Presentations by the Faculty

2026

Sneha. Warrier, Umesh. Mahtani and Smitha. Rao, *"A machine learning approach to analyzing the impact of environmental, social, and governance (ESG) indictors on financial performance,"* in Information Systems for Intelligent Systems - Proceedings of ISBM 2025, Volume 2 , Smart Innovation, Systems and Technologies, 2026.

2025

Best Presenter Award: Dr. Priyambada Tripathi successfully delivered a presentation on *"Singlular Value Decomposition and application"*, for which she was awarded the Best presenter award in Faculty Category during the National Short Term Training Program on "Linear Algebra for Machine Learning" from December 01 to December 05, 2025.

Raj Laxmi, Sachin Kulkarni, Venkatesh Burudi, AV Anupama, Mahipal Ranot *"Annealing temperature dependent phase evolution in heterophasic bismuth vanadate and the effect of isotype-heterojunction on the white light active degradation of methylene blue"* - MRS Advances 10, 2170-2176(2025)

M. Mallikarjuna and Tarachand Amgoth, Senior Member, IEEE

"Energy-Aware Unified Data Collection and Energy Replenishment in Consumer Mobile WSNs" IEEE TRANSACTIONS ON CONSUMER ELECTRONICS, VOL. 71, NO. 2, (MAY 2025)

Internship Highlights

Here are a few of our student internships

*Scholarship * - up to 100%*

Profile-Based Scholarships to Encourage Diversity

*Subject to University Scholarship Policy

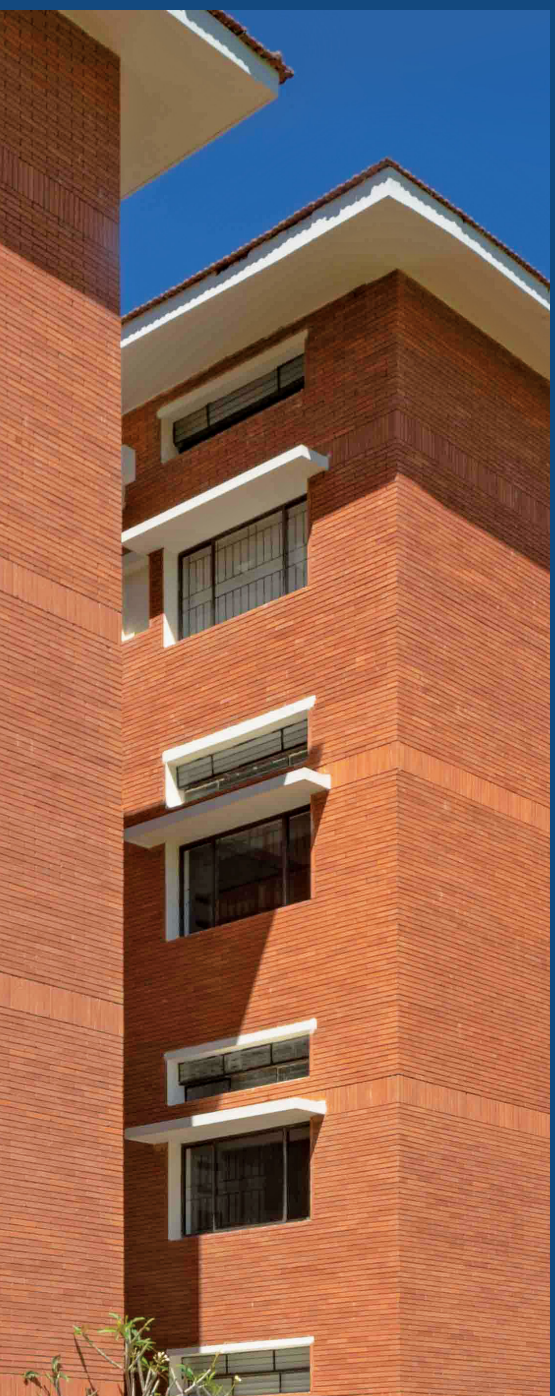
- P. Dayananda Pai Scholarship
- SAT scholarships
- Talent-based scholarships
- Need-based scholarships
- Dean's Honors Scholarship
- Other Categories

28+ crore awarded in scholarships since 2022

Admissions

1. *Submission of the application*
2. *Evaluation of the application based on the academic and non-academic profiles*
3. *Interaction with academic counselor and expert committee*
4. *Portfolio and document evaluation if eligible for a scholarship*
5. *Result notification with the LOS*
6. *Admission confirmation after the payment of acceptance fee and fulfilment of eligibility criteria*

Know More



Message from the Dean

Our BTech in Computer Science & Engineering with honors in Data Science is designed to train professionals for the new age industry. As all industries globally shift towards more data-driven decision making, this program is geared towards training skilled data scientists. At Vidyashilp, we blend theoretical concepts with practical application ideas so that you will mature into a problem solver who can address complex real-world applications.

P S Sastry
Professor & Dean

Apply Now

VIDYASHILP UNIVERSITY

VIDYASHILP SCHOOL OF ENGINEERING & TECHNOLOGY

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Vidyashilp University is established under Karnataka Act No. 20 of 2021